■ Step-by-Step Guide: Creating a Docker Image and Container

This guide explains how to create a Docker image from a Dockerfile and run a container from that image.

1■■ Step 1: Create a Dockerfile

Create a file named Dockerfile in your project directory (e.g., C:\Users\YourName\Desktop\Registration).
Example Dockerfile for a Node.js app:

```
FROM node:18-alpine WORKDIR /usr/src/app COPY package*.json ./ RUN npm install --production COPY . . EXPOSE 3000 CMD ["node", "server.js"]
```

2■■ Step 2: Build the Docker Image

Run the following command in the same directory as your Dockerfile:

```
docker build -t registration:1.0 .
```

• docker build → builds the image • -t registration:1.0 → names it 'registration' with version tag '1.0' • . → means the Dockerfile is in the current directory

3■■ Step 3: Verify the Image Was Created

List your images using:

```
docker images
```

You should see your image in the list.

4■■ Step 4: Run a Container

Create a container from your image using:

```
docker run -d -p 3000:3000 --name registration-container registration:1.0
```

• -d → runs in background (detached) • -p 3000:3000 → maps port 3000 of host to container • --name → gives the container a readable name

5■■ Step 5: Verify Running Containers

List running containers using:

```
docker ps
```

6■■ Step 6: Test the App

Open your browser and visit http://localhost:3000 to view your app running inside Docker.

7■■ Step 7: Stop or Remove the Container

Stop the container:

```
docker stop registration-container
```
Start it again:
```
docker start registration-container
```
Remove the container and image:
```
docker rm registration-container docker rmi registration:1.0
```

■ Summary of Common Commands

docker build -t registration:1.0 . — Build the image
docker images — List images
docker run -d -p 3000:3000 --name registration-container registration:1.0 — Run container
docker ps — List running containers
docker stop registration-container — Stop container
docker rm registration-container — Remove container